Year 13 Further Maths - Differential equations in mechanics

Similar questions - Using v dv/dx

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle P of mass 1.5 kg moves along the x-axis. At time $t = 0$ it passes the origin with speed 4 $\mathrm{m\,s^{-1}}$ in the positive direction. The only force on P is a resistive force of magnitude $\left(v^2+4\right)$ N in the negative x-direction, where v $\mathrm{m\,s^{-1}}$ is the speed.

(a) Find an expression for v in terms of x. **[5]**

(b) Find the distance travelled while the speed drops from 4 to 2 $\mathrm{m\,s^{-1}}$. **[2]**

(c) Explain why P comes to rest after a finite time. **[2]**

(d) Find the total distance travelled by P. **[2]**

2 A body B of mass 2 kg moves along the x-axis. When its speed is 3 $\mathrm{m\,s^{-1}}$ the resistance to its motion is 18 N; the resistance is proportional to v^2 and is the only horizontal force.

(a) Show that B's motion can be modelled by $\dfrac{1}{v}\dfrac{\mathrm{d}v}{\mathrm{d}x} = -1$. **[3]**

(b) Given $v = u$ at $x = 0$, find v in terms of x and u. **[3]**

(c) Describe one feature of the model that is not realistic. **[1]**

(d) Use $W = \int F\,\mathrm{d}x$ to find the work done by the resistance as B moves from O to the point where $x = X$, and show how this relates to the energy of B. **[4]**

3 A particle P of mass 4 kg is at $x = 2$ metres moving in the negative x-direction with speed 3 $\mathrm{m\,s^{-1}}$. An impulse I N s is applied in the positive x-direction so that P moves in the positive direction with speed 10 $\mathrm{m\,s^{-1}}$.

(a) Find I. **[1]**

(b) Subsequently two forces act: $\dfrac{12v^2}{x}$ N in the positive direction and $20v$ N in the negative direction. Show that $\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{av}{x} + b$, stating a and b. **[2]**

(c) Solve to show that $v = \dfrac{5x}{2} + \dfrac{5x^3}{8}$. **[5]**

(d) Find the time for P to move from $x = 2$ to $x = 4$. **[3]**

4 A particle of mass m moves along the x-axis under a force F directed along the axis. Its speed at displacement x is $v = k\sqrt{a^2 - x^2}$, where a and k are constants.

(a) Using the fact that $\dfrac{\mathrm{d}v}{\mathrm{d}x}$ has the same dimensions as $\dfrac{v}{x}$, verify that $F = mv\dfrac{\mathrm{d}v}{\mathrm{d}x}$ is dimensionally consistent. **[2]**

(b) Explain why $[a] = [x]$ and deduce the dimensions of k. **[3]**

(c) Find an expression for F in terms of x, m and k, and describe the motion. **[4]**

Question 1

(a) $1.5v\dfrac{\mathrm{d}v}{\mathrm{d}x} = -(v^2+4)$, so $\displaystyle\int \frac{v}{v^2+4}\mathrm{d}v = -\frac{2}{3}\int \mathrm{d}x$: $\frac{1}{2}\ln(v^2+4) = -\frac{2}{3}x + c$, with $v=4$ at $x=0$: $x = \frac{3}{4}\ln\dfrac{20}{v^2+4}$

(b) $v=2$: $x = \frac{3}{4}\ln\dfrac{20}{8} = 0.687$ m

(c) $1.5\dfrac{\mathrm{d}v}{\mathrm{d}t} = -(v^2+4)$ and the force is at least 4 N even as $v\to 0$: $t = 0.75\left(\tan^{-1}2 - \tan^{-1}\frac{v}{2}\right)$ is finite when $v=0$ ($t = 0.83$ s)

(d) $v=0$: $x = \frac{3}{4}\ln 5 = 1.21$ m

Question 2

(a) $R = kv^2$, $18 = 9k$, $k=2$. $2v\dfrac{\mathrm{d}v}{\mathrm{d}x} = -2v^2$, so $\dfrac{1}{v}\dfrac{\mathrm{d}v}{\mathrm{d}x} = -1$

(b) $\ln v = -x + c$; $v = u\mathrm{e}^{-x}$

(c) v never reaches zero (the body never stops), and the resistance would not fall to zero for tiny speeds in reality

(d) $W = \int_0^X 2v^2\mathrm{d}x = 2u^2\int_0^X \mathrm{e}^{-2x}\mathrm{d}x = u^2\left(1-\mathrm{e}^{-2X}\right)$ (work done *by* B against resistance). With $V = u\mathrm{e}^{-X}$: $W = u^2 - V^2 = \frac{1}{2}(2)\left(u^2-V^2\right)$, the loss of kinetic energy of B

Question 3

(a) $I = 4(10) - 4(-3) = 52$ N s

(b) $4v\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{12v^2}{x} - 20v$, so $\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{3v}{x} - 5$: $a=3$, $b=-5$

(c) IF x^{-3}: $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(vx^{-3}\right) = -5x^{-3}$, $vx^{-3} = \frac{5}{2}x^{-2} + c$, $v = \frac{5}{2}x + cx^3$. $x=2, v=10$: $10 = 5+8c$, $c = \frac{5}{8}$

(d) $t = \int_2^4 \dfrac{\mathrm{d}x}{\frac{5}{2}x + \frac{5}{8}x^3} = 0.0940$ s (partial fractions: $\dfrac{8}{5x(4+x^2)} = \dfrac{2}{5x} - \dfrac{2x}{5(4+x^2)}$, giving $\frac{2}{5}\left[\ln x - \frac{1}{2}\ln(4+x^2)\right]_2^4$)

Question 4

(a) $[F] = \mathrm{MLT}^{-2}$; $\left[mv\dfrac{\mathrm{d}v}{\mathrm{d}x}\right] = \mathrm{M}\cdot\mathrm{LT}^{-1}\cdot\dfrac{\mathrm{LT}^{-1}}{\mathrm{L}} = \mathrm{MLT}^{-2}$ (consistent)

(b) a^2 and x^2 are subtracted so must have the same dimensions: $[a] = \mathrm{L}$. $\mathrm{LT}^{-1} = [k]\mathrm{L}$, so $[k] = \mathrm{T}^{-1}$

(c) $\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{-kx}{\sqrt{a^2-x^2}}$; $F = mk\sqrt{a^2-x^2}\cdot\dfrac{-kx}{\sqrt{a^2-x^2}} = -mk^2x$. Simple harmonic motion about $x=0$ with angular frequency k and amplitude a